

Library Management System

A Streamlit-based Book Issue and Return Application

1. Project Objective

The Library Management System is a Python web application built with Streamlit. It helps a librarian record book issues, view currently issued books, and record book returns. The interface displays member and book information, issue and due dates, and a small library overview.

2. Key Features

- Book issue form with member name, member ID, book title, book ID, issue date, and due date.
- Success message and transaction details after a book is issued.
- Library overview cards for total books, available books, issued books, and registered members.
- Page to view active issued-book records.
- Return-book page that updates the issue status and available-book count.
- Validation for missing fields, invalid dates, and duplicate active book IDs.

3. Tools & Technologies Used

Component	Technology
Programming Language	Python 3
Web Framework	Streamlit
Date Handling	Python datetime
Interface	Streamlit forms, columns, metrics
IDE / Environment	VS Code / Terminal

4. Working / Methodology

The user selects an option from the sidebar: Issue a Book, View Issued Books, or Return a Book. For issuing, the user fills in the member and book details and submits the form. The application validates the details and dates, then creates an issue record and displays a success message with transaction details. The overview metrics are updated. The return page changes the selected record to Returned and increases the available-book count. In this starter version, records are held in Streamlit session state and are not permanently saved to a database.

5. How to Run the Project

1. Install Python 3.
2. Open Command Prompt or the VS Code terminal in the folder containing app.py.
3. Run: **python -m pip install streamlit**
4. Run: **python -m streamlit run app.py**
5. The application opens in your browser, usually at **http://localhost:8501**.

6. Output Screenshot 1 — Book Issue Form



Library Management System

Manage books, members, and borrowing easily



1. Issue a Book

Enter the member and book details below.



Student / Member Name

Ayesha



Member ID

LIB102



Book Title

Introduction to Python



Book ID

BK205



Issue Date

28 Sep 2026





Due Date

12 Oct 2026



Issue Book →

Fig. 1: Library Management System — Book Issue Form

7. Output Screenshot 2 — Book Issued Successfully

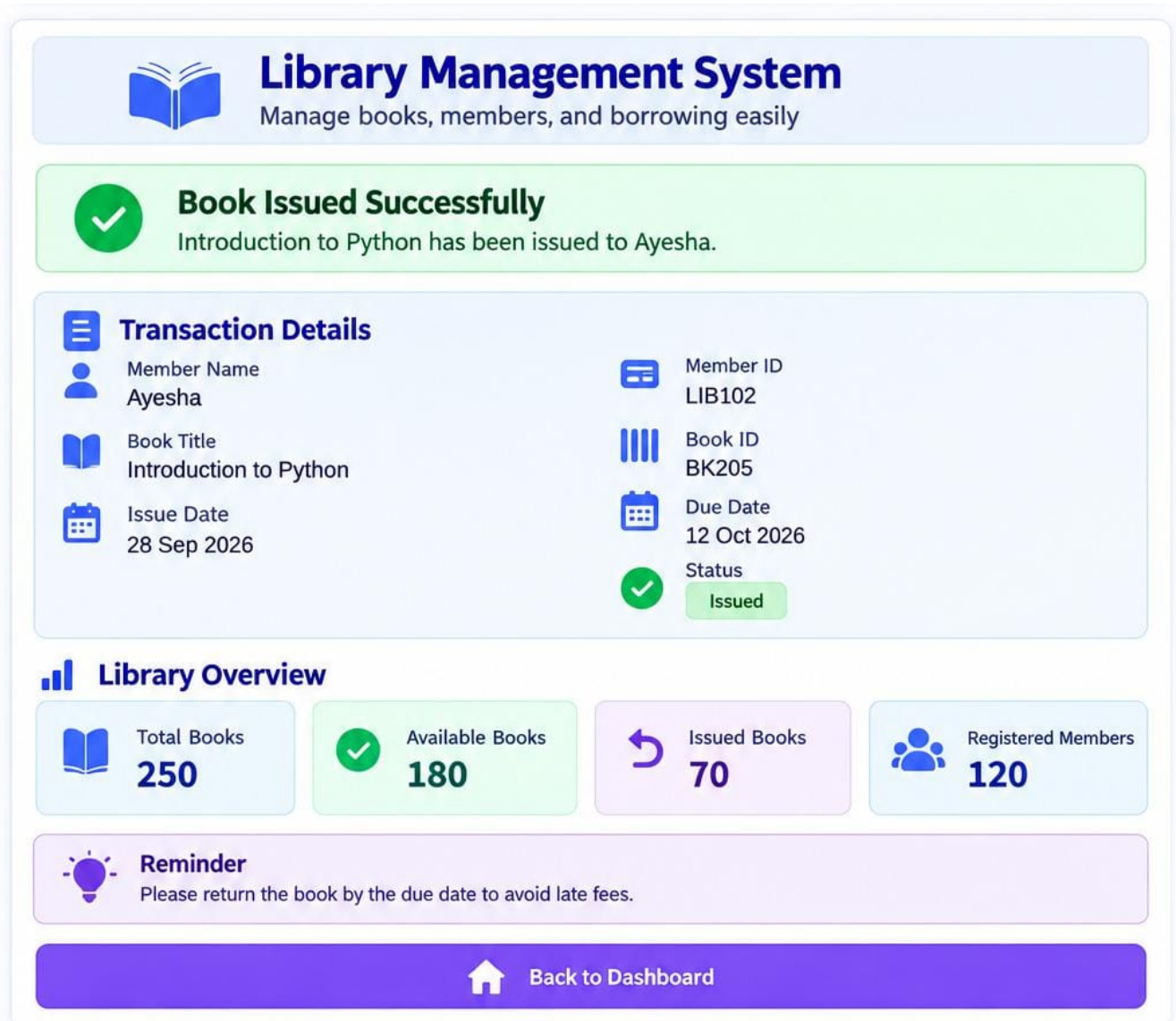


Fig. 2: Library Management System — Book Issued Successfully / Transaction Details

8. Conclusion

The project demonstrates a simple library workflow using Python and Streamlit. It provides a clear interface for issuing and returning books and reviewing active issue records. A future version can connect SQLite or MySQL to save books, members, and transactions permanently, and can add login authentication, search, overdue tracking, and fine calculation.